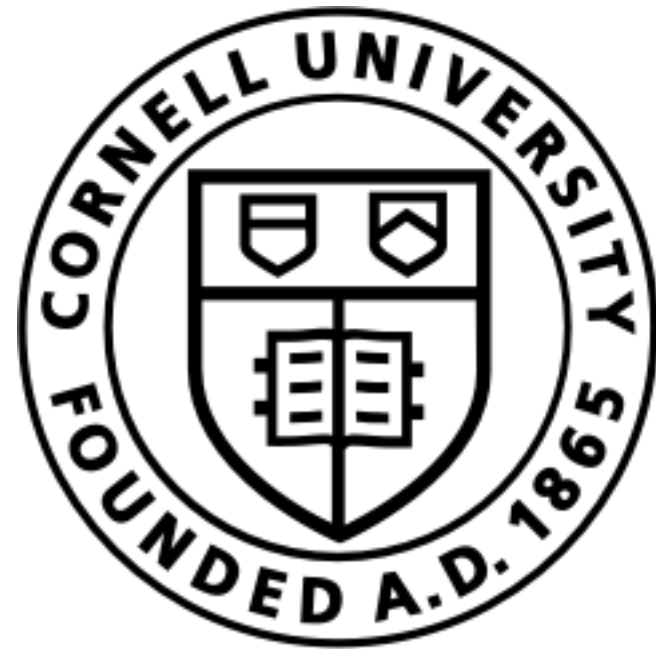




Lecture 2b: Our First Decision Model. Introducing Discrete Choice Models and Random Utility Functions

CHEME 5760 Cornell University

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Objectives: In this lecture we will:

- ▶ **O1:** Introduce observed and unobserved features, the concept of random utility functions (RUMs) and Discrete Choice Models (DCMs)
- ▶ **O2:** Introduce one particular type of DCM: the Logit choice model of McFadden and coworkers
- ▶ **O3:** Revisit the Model S versus Odyssey decision using a Logit choice model and a Bernoulli random variable (Jupyter Notebook example)

Jupyter Notebooks

Repository: <https://github.com/varnerlab/CHEME-5760-Examples-F23.git>

- **Example:** CHEME-5760-L2b-DiscreteChoiceModels.ipynb

Refinement: Utility Functions (Redux)

An individual decision-maker (agent) is given a set of $X_1, \dots, X_n$ objects where each object has m possible features $X_i = \{x_{1,i}, \dots, x_{m,i}\}$. A utility function ranks the agent's preference for combinations of features for each object i :

$$U_i(x_{1,i}, x_{2,i}, \dots, x_{n,i}) = u_i \quad (1)$$

where u_i is a real number called the utility for object i . Utility has units of utils.

- The utility function $U : X \rightarrow \mathbb{R}$ is unique only up to an order-preserving transformation.
- Utility functions are ordinal, i.e., they rank-order bundles but do not measure differences between bundles.

Tesla model S

Link: <https://shop.tesla.com/>



Honda Odyssey

Link: <https://automobiles.honda.com/odyssey>

Scoring system



Features	Exponents	Tesla	Honda
Sustainability	0.2	5	3
Affordability	0.1	2	4
Styling	0.05	5	3
Usefulness	0.3	2	5
Cost of Ownership	0.1	4	2
Performance	0.05	5	1
Safety	0.2	5	5
Build quality	?	?	?
Feature ...	?	?	?

unobserved features

Concept: Random Utility Functions (RUMs)

For an individual, both observable and unobservable features can influence decisions. Utility functions for individuals that consider both observable and unobservable factors take the form:

$$U_{ij} = V_{ij} + \epsilon_{ij} \quad (2)$$

The term U_{ij} is the utility of alternative j for individual i , V_{ij} is the *deterministic* component of the utility, i.e., the utility associated with the observable features, and ϵ_{ij} is the random component of the utility (error model).

Concept: Logit Choice Probability

If we assume the random component of the utility is *independently and identically distributed* (IID) across J alternatives, and is **Gumbel distributed**, then the probability that individual i chooses alternative j is given by the **logit** model:

$$P_{ij} = \frac{e^{V_{ij}/\mu}}{\sum_{k=1}^J e^{V_{ik}/\mu}} \quad j = 1, \dots, J \quad (3)$$

where P_{ij} is the probability that individual i chooses alternative j , V_{ij} is the deterministic component of the utility, and μ is a scale parameter.

A Bernoulli random variable X models a binary outcome: either 1 or 0, where 1 occurs with probability p and 0 occurs with probability $1 - p$. The probability mass function (pmf) of the Bernoulli random variable X is:

$$p_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases} \quad (4)$$

where $0 < p < 1$ is called the Bernoulli parameter. The expectation a Bernoulli random variable equals:

$$\mathbb{E}[X] = p \quad (5)$$

while the variance $\text{Var}(X)$ equals:

$$\text{Var}[X] = p(1 - p) \quad (6)$$

Bernoulli random variables model many binary events: coin flips (H or T), binary bits (1 or 0), true or false, yes or no, present or absent, etc.

[switch to the DCM notebook example]

Next week we start to consider reality: We revisit the ideas from the last few lectures in light of budget and other types of constraints.

What are we doing in the discussion section on Friday? We are going to explore a discrete choice travel model dataset from Dresden, Germany

- [Sven Müller, Stefan Tcharaktschiew, Knut Haase \(2008\) Travel-to-school mode choice modeling and patterns of school choice in urban areas. Journal of Transport Geography, 16: 342-357, https://doi.org/10.1016/j.jtrangeo.2007.12.004](https://doi.org/10.1016/j.jtrangeo.2007.12.004)

Today?